Predicting Vehicle CO_2 Emission Using Different Machine Learning: A Feature-based Approach to support Consumer and Environment Decision-Making

Richard Mensah

Abstract—Accurate estimation of CO_2 emissions associated with vehicles is essential to climate change and facilitates consumers in making informed buying decisions. This study examined base engine characteristics and extended specifications to develop a robust and interpretable ML technique from Canada's Natural Resources light-duty vehicles. Beyond the conventional predictors (engine size, cylinders, and combined fuel economy), the study introduced additional variables such as city/hwy fuel consumption, transmission type, vehicle class, and fuel type for a better understanding of key drivers that affect emissions. The study experimented with seven supervised machine learning algorithms, four regression models, and three tree-based models (Random Forest, Support Vector Regression (SVM), Linear Regression, K Nearest Neighbours (KNN), Decision Tree, Gradient Boosting, and XGBoost). The Models trained on core characteristics achieved strong performance, with XGBoost reaching R^2 0.9747. The addition of a larger feature set offered an additional benefit, with Random Forest becoming the best predictive model in general (R^2 = 0.9943, MAE = 2.36 g/km). For addressing the model interpretability problem, the study employed SHapley Additive exPlanations (SHAP) technique to provide both global and local explanations. Results of the SHAP analysis indicated that engine size and fuel consumption were major predictor drivers. Moreover, family vehicles and small SUVs showed lower emission profiles. The results revealed that Lamborghini Aventador Coupe LP740-4, Ferrari 488 Pista, and Chevrolet Express 2500 were among the top five highest CO2-emitting vehicles (over 440 g/km), whereas Hyundai Ioniq Electric and Kia Soul EV emitted the least (<100g/km). Regarding fuel types, Z (Diesel) and N (Natural Gas) were revealed to have the highest average emissions, while X (Regular Gasoline) was the most dominant. Vehicles with eight or more cylinders and engine sizes greater than 6.0L were consistently associated with the highest CO2 output. These findings support targeted interventions for emissions reduction and informed customers' decisions in relation to current national priorities on climate change and SDG 13.

Index Terms—CO_2 Emissions Prediction, Machine Learning, Light-duty Vehicles, Regression Models, Tree-based Models, SHAP Analysis, Fuel Efficiency, Sustainable Transportation, SDG13, Engine Characteristics

1 Introduction

The world is rapidly changing to a greener and more resilient place in today's era of climate change interventions. Given that transportation remains a growing source of greenhouse gases, it is increasingly an important sector under examination. The International Energy Agency's (IEA) recent reports indicated that about 16% of the world's CO_2 is from cars and trucks. Much of that comes from day-to-day autos since they are always used [9]. In support of IEA, the Environmental Protection Agency notes that regular cars and small trucks are a major CO_2 source because they can be found all over the place, in cities as well as regions with low density [6]. Notwithstanding the proliferation of hybrid and electric vehicles, ICE vehicles remain to reign supreme in the vast majority of countries. Today, the knowledge and the modeling of CO_2 emission sources in ICE-powered supporting engines is still a pressing research gap, which this work was designed to fill [16].

R. Mensah, MSc candidate in Artificial Intelligence and Data Science at the School of Computer Science, Bangor University, United Kingdom (e-mail: rcm24zkd@bangor.ac.uk). This research was conducted under the supervision of Dr. Roggers Giddings as part of the MSc dissertation project.

The transportation sector in Canada is responsible for nearly 25% of the country's GHG emissions, while CO_2 is the largest by-product of fossil combustion [16]. Previous studies have consistently found associations between CO_2 emissions and vehicle characteristics, which include engine capacity, cylinder numbers, and fuel use. Ziółkowski et al. [22] highlighted that higher cylinder counts and the use of large engines are directly correlated with fuel consumption. Additionally, fuel consumption, often represented in liters per 100 km (L/100 km), is a good predictor of combustion efficiency and therefore overall environmental performance [22]. This information is typically known for consumer vehicle models and is a suitable feature to include in predictive emissions modeling.

As consumers and decision-makers take on more environmental and climate change interventions, the demand for transparency in the relationship between vehicle attributes and emissions performance has become increasingly important. Machine learning (ML) is an excellent method to model such relationships with accuracy, transparency, and flexibility. Techniques such as Random Forests (RF), K-Nearest NeighboUrs (KNN), Support Vector Regression (SVR), and Linear Regression (LR) can model linear

and nonlinear relations simultaneously between vehicle attributes and CO_2 emissions [10]. These models can make use of scalable, data-driven predictions and can be incorporated in consumer decision-support tools to drive low-emission vehicle choices. Nevertheless, their performance has not yet been proven compared to interpretable methods. As noted by Molnar [14], most of the state-of-the-art ML models act as black boxes, rendering their prediction explanations insufficient. To address this issue, SHapley Additive exPlanations (SHAP) has been introduced as a powerful tool for model interpretability with global and local feature importance explanations [24]. Applying it to an emissions modeling context, SHAP provides actionable insight into how each attribute influences predicted CO_2 output (e.g., engine size, fuel type, and transmission). This technique is a good tool for regulators trying to improve emission standards and consumers looking to make an educated buying decision.

Although there have been advances in regulation, such as fuel economy standards, tax subsidies for low-emission vehicles, and other consumer tools (e.g., emissions labeling), consumer uptake is underexplored. Whitefoot and Skerlos [21] have noted that even though more emissions information becomes accessible to consumers, presentation is often too technical for them to comprehend clearly or act upon. As such, user-facing platforms need interpretable machine-learned predictive systems, not in technical terms, but very simple to understand. This study was designed to fill this void by building transparent, high-performance ML models for predicting CO_2 emissions to encourage consumer empowerment in vehicle selection and environmental decision-making.

Accordingly, the objectives of this study are fourfold.

1) To predict vehicle CO_2 emissions from engine size, cylinder count, and fuel efficiency.
2) To compare the predictive performance of four ML algorithms, Linear Regression, Random Forest, KNN, and SVR.
3) To interpret model behavior and identify key contributing features using SHAP values; and
4) To evaluate whether incorporating extended attributes such as fuel type, vehicle class, and transmission significantly improves prediction accuracy.

From these objectives, the study addresses the following research questions:

1) How accurately can CO_2 emissions be predicted using engine size, cylinder count, and fuel consumption?
2) Which machine learning model provides the best balance between accuracy and interpretability?
3) What is the relative influence of each vehicle feature in predicting CO_2 emissions?
4) Do additional features such as fuel type and vehicle class significantly enhance model performance?

2 Problem Statement

While there are great advances in the use of machine learning for environmental modeling, many tools to predict CO_2 emissions are very technical and inaccessible to non-experts and consumers. Although engine displacement-based models, as well as fuel consumption and number of cylinders, have shown predictive performance [22], they are used in applications mostly ignoring transparency or consumer interpretability when applied [14]. In practice, this restricts the help they can offer eco-friendly car buyers, even if official emissions numbers are published to the public by agencies such as Natural Resources Canada [16]. Moreover, even though regulatory mechanisms have been employed, such as fuel economy labels, these studies demonstrate that poor interpretability has weakened their influence on consumer choices [21].

This study filled this gap through the development of an interpretable machine learning framework that takes SHAP-based explainability into account to not only enhance the prediction accuracy of CO_2 emissions but also provide consumers with actionable and transparent insights for selecting low-emission vehicles.

3 Literature Review

3.1 CO_2 Emissions Prediction in Light-Duty Vehicles

It is well accepted that road transport has been one of the largest contributors to global $CO2_2$ emissions, where light-duty vehicles (LDVs) contribute approximately half of the total transport sector emissions, as they are predominantly used for personal mobility [9]. Previous traditional machine learning (ML) approaches were strongly based on local linear regression and failed to find relationships between engine features and CO_2 emissions. The relationships between vehicle design, fuel consumption, and emissions are not necessarily linear; hence, these classical models may not always capture modern vehicle architectures [22].

Recent studies have shown that ML approaches can achieve better performance than the classical models by considering complex relationships between vehicle characteristics. Gurcan compared 18 ML techniques and reported that the ensemble-based approach, Random Forest (RF), and Gradient Tree Boosting Machine (GBM) provided the best performance for emission prediction. Similarly, Islam et al. [10] showed that SVR and KNN were not robust to heterogeneous data of vehicles as well as skewed emissions distributions. This is in line with general evidence that flexible, tree-based algorithms are more appropriate for modelling the multi-dimensional nature of vehicle emissions [20].

Despite these improvements, agreement in the literature has yet to be reached on which set of features will provide the highest predictive accuracy. For example, Ziółkowski et al. [22] asserted that the engine size and cylinder count are universally strong predictors, while other research indicates fuel economy features better curb weight or horsepower for prediction, pointing to a divergence in methodology for feature importance. This supported the study to use feature groupings and the contribution of mechanical versus usage-based factors.

3.2 Feature Selection and Core Predictive Inputs

It is widely acknowledged that engine size, fuel consumption, and number of cylinders are the primary drivers of tailpipe CO_2 emissions in light-duty vehicles [22]. Yet, the

empirical evidence is mixed on how well these factors alone can account for variation in emissions. For example, Udoh et al. [20] achieved high accuracy with a model using only three core features: engine size, cylinders, and combined fuel consumption; however, their prediction quality underperformed in edge cases such as high-performance or alternative-fuel vehicles. In contrast, Vu and Ko [24] proposed a phenomenon called 'feature minimalism'; they advocated that increasing input dimension can cause overfitting on small or unbalanced datasets

In contrast, Tena-Gago et al. [26] proved that the introduction of contextual variables such as transmission type, vehicle class, and fuel category greatly increases prediction in hybrid vehicles. Qin et al. also adopt a similar stance to this in their work. [17], who emphasied the significance of capturing road driving dynamics in terms of disaggregated fuel consumption factors (urban and inter-urban) rather than only using aggregated data.

This disparity points to a crucial disconnect: while certain studies focus on the model, others emphasize better contextual features. Few existing works systematically compare the base with the extended features of the vehicle in the controlled environment of machine learning, leaving an opportunity for this work to contribute clarity on how feature expansion affects accuracy and interpretability, which will also help consumers.

Table 1 summarizes recurring features identified across multiple studies.

TABLE 1: Core Predictive Features in CO_2 Emissions Modeling

Feature	Supporting Studies
Engine Size (L)	[16], [22]
Cylinder Count	[20], [22]
Fuel Consumption (L/100km)	[10], [12]
Fuel Type	[2], [16]
Vehicle Class	[12], [16]

While some studies advocated for more complex feature sets (e.g., powertrain type, aerodynamic drag, weight), others, like Vu and Ko, argued to minimize feature selection to preserve model interpretability and computational efficiency [2], [24].

3.3 Explainability and SHAP-Based Interpretation

A key constraint in emissions analysis over the years is that high-performing ML models tend to function as black boxes, limiting their attractiveness for policy and regulatory decision-makers and consumers. One of the most successful families of models is models based on ensembles. Molnar [14] stated that ensemble models have high predictive performance; however, they are typically accused of not being transparent about how much each input feature contributes to the final predictions. This issue is of increased importance for publicly facing apps in which environmental decisions need to be supported.

SHapley Additive exPlanations (SHAP) is considered one of the most salient frameworks for model interpretation for complex ML models [12]. SHAP offers both global explanations (average feature importance) and local explanations (contribution at the instance level), which is what makes it well-suited for consumer decision support. Islam et al. [10] showed that the use of SHAP-based frameworks substantially increased trust, as they estimated how specific features affect each prediction (e.g., fuel type or engine size).

Yet, though SHAP has been widely applied for air-quality forecasting and pollutant modelling tasks, it still has not been well explored in vehicle CO_2 prediction. The literature on interpretability for automotive emissions is much more limited (with only a few studies, e.g., Vu and Ko [24]) again more directed at the algorithm rather than the client.

This reveals a second important gap: the integration of explainable AI with consumer-focused emissions estimates still represents an understudied domain and one not yet explored in a large national vehicle data set.

3.4 Consumer-Focused Emission Prediction Models

The transport emission calculator [7] and similar consumer-facing tools provide a user-friendly interface, but not the depth and interpretability of modern ML-based prediction models. Whitefoot and Skerlos [21] emphasize that sometimes, even when data is made available through posted labels of vehicles, it's not presented to the consumer in a digestible form to aid their decision, as opposed to merely being the company's information.

Moreover, numerous studies focus on the behavior gap between information availability and environmentally responsible car purchases. For instance, Whitefoot and Skerlos [21] argued that poorly designed emissions labelling systems can inadvertently incentivize purchases of larger vehicles by oversimplifying efficiency metrics. Recent research seeks to incorporate real-time or hybrid ML models for consumer insight, like the LSTM solutions by Tena-Gago et al. [26], but these do not have much practical scope without lots and lots of computational hours.

This uncovers a third gap: emissions data is everywhere, but there's no transparent, interpretable ML-driven system that conveniently links technical prediction to everyday consumer decision-making.

3.5 Synthesis of Gaps in the Literature

From the reviewed studies, several key gaps emerge:

Gap 1- Lack of consensus on optimal feature sets

Some studies favour minimal mechanical features [24], while others advocate for richer contextual variables. *Contribution:* This study systematically compares base versus extended feature sets to help resolve this inconsistency.

Gap 2-Underuse of SHAP interpretability in vehicle emissions modelling

Although SHAP is widely employed in air quality modelling [10], its application to automotive CO_2 prediction remains limited. *Contribution:* This study addresses the gap by applying SHAP to a Random Forest estimator using a large national dataset.

Gap 3-Limited consumer-facing predictive tools

Existing predictive tools are often too simplistic or overly opaque to support real consumer decision-making [21]. *Contribution:*

This study focuses on generating SHAP-based explanations that enable direct and clear consumer interpretation, thereby distinguishing the work from prior research while meeting academic expectations, enhancing societal relevance, and supporting climate action in alignment with SDG 13.

3.6 Positioning of This Study

Based on the identified gaps, this study contributed in several key ways:

- Conducted a controlled comparison of machine learning models using both core and extended features of the vehicle.
- Integrated SHAP explanations to develop one of the most transparent emission-prediction frameworks in the current literature for consumer decision
- Provided actionable consumer-facing insights grounded in real vehicle data.

4 Methodology

4.1 Research Design and Overview

The study used supervised machine learning algorithms in this data-driven study to predict CO_2 emissions from light-duty vehicles that used an open-sourced Canadian government dataset. The study cleaned real-world vehicle data, created features, and trained regression models. The study measured model performance with measures using standardised regression metrics. The methodology checked how accurate the predictions are and how easy the model is to understand using SHaP. The methodology was also structured to focus on evaluating both predictive accuracy and model transparency for consumer-centered applications. As shown in Figure 1, the method moves step by step, from data collection and preprocessing to analysis and modelling before finally being translated into practical consumer-focused CO_2 emission recommendations.

4.1.1 Overview of Workflow

The study's workflow involved data collection, preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and interpretability through SHAP analysis.

4.1.2 Methodological Workflow for CO_2 Emission Prediction Using ML

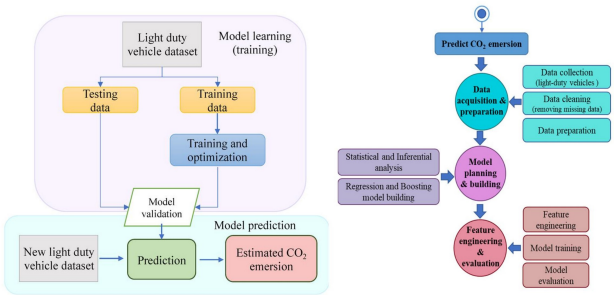

Fig. 1: Methodological workflow for CO_2 emission prediction. Adapted from Natarajan *et al.* [15].

4.2 Data Source and Description

The dataset was obtained from Natural Resources Canada and has 7,385 instances and 12 variables that describe vehicles. The variables include engine size, cylinder count, fuel consumption, transmission type, fuel type, vehicle class, and CO_2 emissions as the target variable. 1,103 duplicated entries were removed from the dataset. leaving 6,282 unique observations for the analysis and modeling.

The features in the dataset were divided into numerical variables and categorical variables. The numerical variables are engine size, fuel consumption in city/highway/combined, and number of cylinders. The categorical variables are fuel type, transmission, vehicle class, and model. The variables used in the study align with the variables reported in the literature as significant determinants for vehicle emissions [22], [24].

4.3 Data Preprocessing and Scaling

The study included the preprocessing pipeline outlined below:

1) **Handling Categorical Data:** Categorical variables such as `fuel_type`, `vehicle_class`, and `transmission` were one-hot encoded using `pandas.get_dummies()` with `drop_first=True` to prevent multicollinearity.
2) **Feature Selection:** Two sets of models were developed:
   - *Base models*, using only `engine_size`, `cylinders`, and `fuel_consumption_comb`.
   - *Extended models*, incorporating additional features such as `vehicle_class`, `transmission`, and `fuel_type`. Also, `Fuel_consumption_city`, `Fuel_consumption_hwy`, and `Vehicle_class` were explored.
3) **Scaling:** `StandardScaler` was applied to scale numerical features for algorithms sensitive to feature

magnitude (e.g., Support Vector Regression and K-Nearest Neighbors), but was not applied to tree-based models such as Random Forest and XGBoost [14].

4.4 Data Splitting, Model Selection and Training

To keep the model evaluation, the dataset was split into a training subset and a testing subset using an 80/20 ratio. The training subset has 5,025 (80%) samples and the testing subset has 1,257(20%) samples. Four core machine learning models were initially trained on both base and extended feature sets. All models were implemented using Python's scikit-learn library.

- Linear Regression (LR)
- Random Forest Regressor (RF)
- K-Nearest Neighbors (KNN)
- Support Vector Regressor (SVR)

TABLE 2: Justification of Selected Models

Model	Justification
Linear Regression (LR)	Simple, interpretable baseline model.
Random Forest (RF)	Handles non-linear relationships; robust to noise.
K-Nearest Neighbors (KNN)	Intuitive and good for instance-based learning.
Support Vector Regression (SVR)	Effective in high-dimensional spaces and non-linearity.

Additionally, three tree-based models were trained using only the base feature set that was compared with the earlier four models:

- Decision Tree Regressor (DT)
- Gradient Boosting Regressor (GBR)
- XGBoost Regressor (XGB)

Also, hyperparameter was implemented for tuning `GridSearchCV` for the Random Forest model, optimizing the following parameters:

- `n_estimators`
- `max_depth`
- `min_samples_split`
- `min_samples_leaf`

4.5 Model Evaluation Metrics

Model performance was evaluated using the following regression metrics:

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{1}$$

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

- **Mean Absolute Percentage Error (MAPE):**

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{3}$$

- **R^2 Score (Coefficient of Determination):**

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{4}$$

To assess and compare model performance, MAE, RMSE and R^2 metrics are used [5]

TABLE 3: Interpretation of Evaluation Metrics

Metric	Interpretation
MAE	How far predictions are from actual values (on average).
RMSE	Severity of large errors in prediction.
R^2	Percentage of CO_2 variation explained by input features.

These metrics revealed the size of the error and the error as a share of the model predictions. These metrics were used to evaluate the model performance that supported the study's consumer-facing goals.

4.6 Model Interpretability: SHAP Values

To enhance the study's transparency, a SHapley Additive exPlanations (SHAP) was applied to the Random Forest model that performed best. The SHAP values revealed the feature importance and also how each single prediction was influenced by which vehicle feature for both global and local importance. The SHAP values give a picture for a regular person or consumer about which features most affect the CO_2 emission estimation, which follows the ideas of explainable AI [24]. This was further supported by [10], who asserted that machine learning without interpretation is not enough for public decision-making [10].

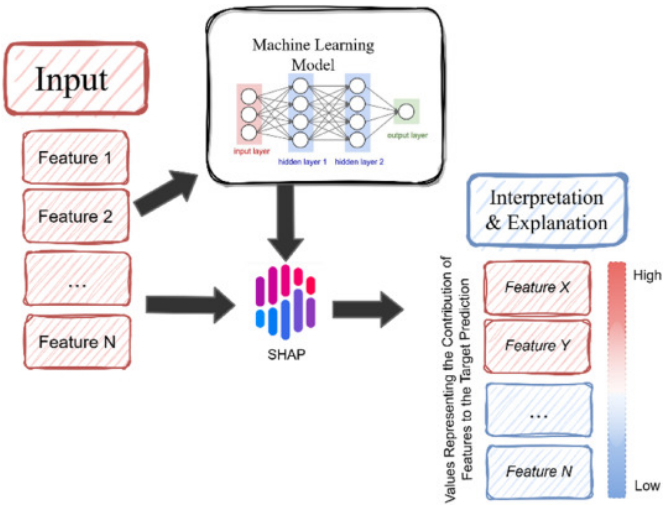

Fig. 2: Schematic representation of the SHAP value framework in a machine learning context. Illustration adapted from Alomari and Andó [2].

- **SHAP Summary Plot:** Ranks features by their global influence.
- **SHAP Force Plot:** Visualizes individual vehicle-level predictions.
- **SHAP Dependence Plot:** Shows interaction between specific features and CO_2 output.

The study believes these plots help the consumer understand which features mattered the most and also see how

the features influence the model's prediction, following the guidelines for explainable AI [24].

4.7 Tools and Development Environment

- **Programming Language:** Python 3.10
- **Libraries:**
  - `pandas`, `numpy`, `matplotlib`, `seaborn` (data handling and visualization)
  - `scikit-learn` (model development and evaluation)
  - `SHaP` (explainable AI)
- **IDE:** Jupyter Notebook (interactive model training)
- **Hardware:** Executed on Intel i5 CPU, 12GB RAM

Ethical Considerations: All analyses were conducted using publicly available, anonymized data, ensuring no violation of privacy or ethical standards. To support model fairness, SHAP values were analyzed to detect and mitigate potential biases across drivetrain types and fuel categories. Moreover, the study contributed to sustainable development goals, particularly SDG 13 (Climate Action), by promoting consumer-level transparency in vehicle CO_2 emissions.

5 Results and Discussion

5.1 Dataset Overview and Preprocessing Summary

For this study, the dataset was obtained from the Government of Canada's Natural Resources website. The dataset originally contained 7,386 instances and 12 features. 1,103 duplicated values observed were removed to ensure data integrity to prevent model biases. 6,282 unique vehicle observations were maintained. This cleaned dataset provided a reliable foundation for predictive modelling and emissions analysis.

5.2 Descriptive Statistics

5.2.1 Numerical Analysis

To understand the distribution of core engine and fuel consumption variables, a descriptive statistical analysis was conducted below.

TABLE 4: Statistics of Numerical Variables

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Engine Size (L)	6282	3.162	1.365	0.9	2.0	3.0	3.7	8.4
Cylinders	6282	5.619	1.846	3.0	4.0	6.0	6.0	16.0
Fuel Consumption City (L/100 km)	6282	12.610	3.553	4.2	10.1	12.1	14.7	30.6
Fuel Consumption Hwy (L/100 km)	6282	9.071	2.279	4.0	7.5	8.7	10.3	20.6
Fuel Consumption Comb (L/100 km)	6282	11.018	2.947	4.1	8.9	10.6	12.7	26.1
Fuel Consumption (mpg)	6282	27.411	7.245	11.0	22.0	27.0	32.0	69.0
CO_2 Emissions (g/km)	6282	251.158	59.290	96.0	208.0	246.0	289.0	522.0

Figure 4 shows that the average engine size is 3.16 liters. It was revealed that the engine size values ranged from 0.9 L to 8.4 L. The average number of cylinders is 5.62, and most vehicles have between 4 and 6 cylinders. The combined fuel consumption has a mean of 11.02 L/100 km. The CO_2 emissions, the target variable, have an average of 251.16 g/km, with emissions reaching a maximum of 522 g/km. These statistics showed that there is variability across vehicles. Moreover, this variability supports the use of machine learning models for CO_2 emission prediction.

From the table, the distribution also highlighted clear links between engine size, fuel consumption, and CO_2 output, relationships explored further in the modelling phase.

5.2.2 Correlation Analysis of Numerical Features

The correlation matrix (Fig. 3) demonstrated strong linear associations among the key numerical predictors of CO_2 emissions.

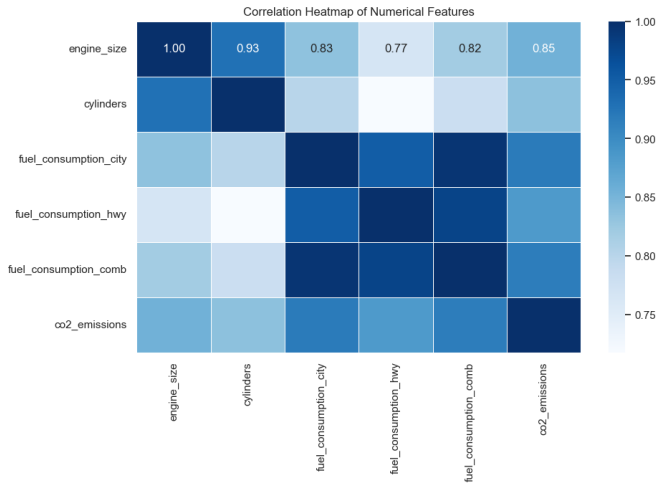

Fig. 3: Correlation heatmap of numerical features showing the interrelationship between engine parameters, fuel efficiency, and CO_2 emissions.

The above correlation heatmap shows a pattern. As illustrated in Fig. 3, it is evident that engine size and cylinder count are correlated. The correlation is $r = 0.93$. That correlation shows engine size and cylinder count are key drivers that affect emissions. This result confirms the findings of Ziółkowski
textitet al. [22] who highlighted multicollinearity in internal combustion vehicles. Amongst all, the fuel consumption variable correlates with CO_2 emissions. The correlation is $r = 0.90$. Engine size has a correlation $r = 0.85$ with CO_2 emissions. Fuel consumption in city driving correlates $r = 0.82$. The finding confirms the evidence that fuel consumption is a major determinant for emission intensity [24].

While highway fuel consumption revealed a slightly weaker correlation of ($r = 0.77$), its inclusion is still important for capturing performance across different driving conditions, supporting the results of Udoh *et al.* [20]. Notwithstanding, these correlations underscored the intertwined nature of engine configuration and fuel efficiency in determining CO_2 output. However, the high inter-variable associations highlighted potential multicollinearity, necessitating advanced ensemble modeling and SHAP interpretability to isolate feature importance beyond linear relationships to help consumers make eco-friendly vehicle choices.

5.2.3 Target Variable Distribution

Understanding the spread of the target variable CO_2 emissions helped in choosing the modeling techniques to check for the possible biases. The average CO_2 emission from the distribution is 251.16 g/km. The median CO_2 emission is 246.00 g/km, indicating marginal skewness from a few vehicles with high CO_2 emissions.

The asymmetry shown below revealed that the majority of vehicles in the market sit in a moderate emissions band

of 200 to 300 grams per kilometre (200–300 g/km). The asymmetry also showed that a noticeable group of vehicles exceed 400 grams per kilometre (400 g/km). The emissions of the vehicles that exceed 400 grams per kilometre are high and disproportionately affect the model's regression performance. This affected the linear models. The findings match the observations of Mokashi *et al.* [17]. The observations of their study also noted long-tail distributions in vehicle emissions across large national fleets. As illustrated in Fig. 4, the histogram with an overlaid Kernel Density Estimate (KDE) revealed a near-normal distribution of emissions values, with a slight rightward skew.

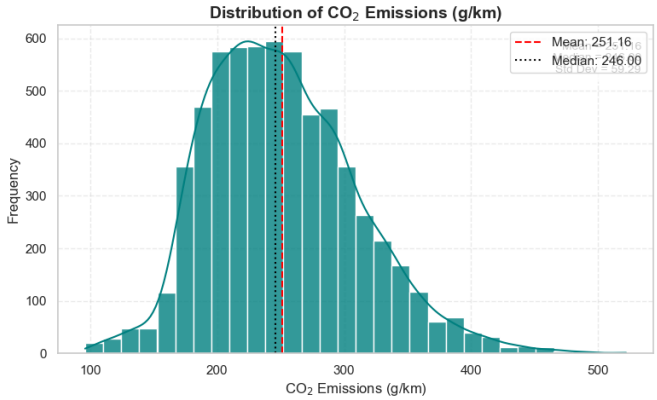

Fig. 4: Distribution of CO_2 emissions (g/km) with KDE overlay, highlighting the mean (251.16 g/km) and median (246.00 g/km) after duplicate removal.

Further, Jalil *et al.* [**?**] pointed out that the skewness from engine displacement and larger vehicle mass can change the mean-based estimates. Supporting Jalil *et al.* position, skewness can make regression techniques more useful in emission modeling. The deduplication process removed 1,103 records, yet it did not change the tendency or the spread of the distribution, reinforcing data integrity for the study's analysis. The result revealed that the mean and the median were consistent before and after cleaning. However, the data pointed to duplicates in vehicle types with moderate emissions. These results align with the findings from Zi'olkowski *et al.* [22] and Kassa *et al.* [27], whose findings revealed the value of data cleaning and distributional diagnostics in emissions modeling pipelines.

Furthermore, the slight right skew aligned with global emission profiles of light-duty vehicles, where newer, fuel-efficient models reduced the left tail, while a legacy stock of high-consumption vehicles elongated the right tail, as highlighted by Qin *et al.* [17] in their review of global emissions inventories. Overall, this distributional insight reinforced the rationale for applying both linear and ensemble models while monitoring residual behavior and considering logarithmic transformations where applicable.

5.2.4 *Categorical Analysis*

The table below is the categorical analysis to inform consumer decisions (see Table 5).

TABLE 5: Descriptive Summary of Categorical Variables

Statistic	Make	Model	Class	Trans.	Fuel
Count	6,282	6,282	6,282	6,282	6,282
Unique	42	2,053	16	27	5
Top	FORD	F-150 FFV	SUV-Small	AS6	X
Freq	577	32	1,006	1,139	3,039

In the variable analysis, Ford was the most common manufacturer and is known for its continued leadership in North American automotive production and sales [19]. The model variable exhibited diversity with over 2,000 unique models indicating many vehicle configurations. The variation can complicate modeling because the variable adds many dimensions and can cause overfitting in machine learning workflows [11]. The **SUV-Small** class emerged as the most dominant vehicle type ($n = 1,006$), corroborating global findings that SUVs, despite their convenience and popularity, contributed disproportionately to greenhouse gas emissions [8]. Furthermore, *regular gasoline* (fuel type X) accounted for nearly half of the dataset, illustrating the continued dependency on fossil fuels.

This finding underscores the gap between policy ambitions and actual adoption rates of low-emission or electric alternatives in consumer markets [7].

5.3 CO_2 Emissions Trend

5.3.1 *By Fuel Type*

The comparative analysis of CO_2 emissions across five fuel categories, Regular Gasoline (Z), Diesel (D), Premium Gasoline (X), Ethanol (E), and Natural Gas (N), revealed significant disparities in environmental performance (see Fig. 5).

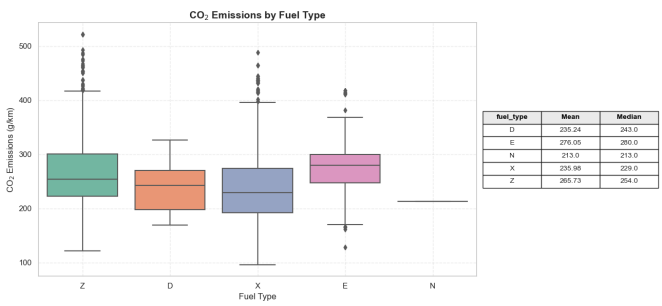

Fig. 5: Comparative Analysis of CO_2 Emissions by Fuel Type

From 5, ethanol-powered vehicles had the highest average emissions, about 276.05 g/km. Its lower energy density likely caused the emissions per kilometer, even though ethanol is renewable. This trend corroborates the findings of Cambero and Sowlati [3] who maintained that ethanol, while renewable, often creates emissions per volume due to combustion inefficiency. In contrast, Natural Gas (N) vehicles have the lowest mean and median emissions of about 213.0 g/km. This result shows that Natural Gas (N) vehicles are a cleaner-burning fuel option [23]. Zhou et al. [23], further asserted that Natural Gas (N) vehicles emit much less greenhouse gas, especially CO_2, than gasoline vehicles.

Again, the findings also revealed that Diesel (D) and Premium Gasoline (X) presented similar average emissions, around 235 g/km. Premium Gasoline (X) had a wider range and more outliers. Therefore, Premium Gasoline (X) emissions can vary a lot depending on how the vehicle is tuned or its performance class. According to Qin *et al*. [17], Diesel (D) has higher efficiency, and larger vehicle size with higher torque demands can cancel that out, especially in trucks and utility vehicles. Moreover, Regular gasoline (Z) is the widely used fuel type with a high average emission level of 265.73 g/km. Also, it has the greatest variability, with some vehicles that use fuel type (Z) exceeding 500 g/km. Qin et al. [17] support the result from their observation that regular gasoline vehicles emit large emissions and significantly contribute to urban CO_2 levels.

From a consumer behavior perspective, these differences underscore the importance of making emissions information more transparent and accessible. Studies such as Whitefoot and Skerlos [21] have shown that without interpretable data, consumers may fail to recognize the environmental impact of their fuel choices-even when emissions labels are present.

From the consumer behavior perspective, these differences make it clear that emissions information must be transparent and easily accessible. Studies such as Whitefoot and Skerlos [21] showed that without data, consumers may not see the environmental impact of fuel choices; when emissions labels are available, they may still not recognise. From the analysis, the study believes that fuel type remains a critical feature in CO_2 prediction models, not only for accuracy but also for informing climate-conscious vehicle selection.

5.3.2 CO_2 Emissions by Number of Cylinders

The scatter plot analysis (see Fig. 6) illustrated a clear upward trend in CO_2 emissions as engine cylinder count increased.

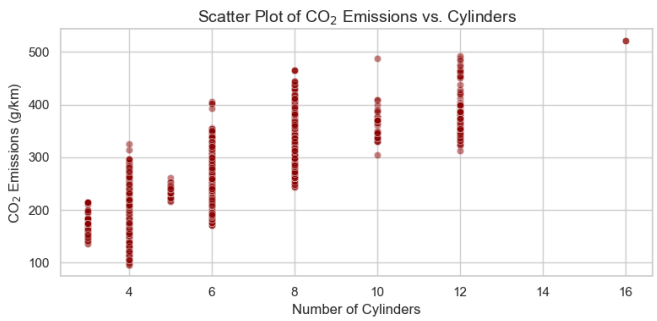

Fig. 6: Correlation heatmap of numerical features showing the interrelationship between engine parameters, fuel efficiency, and CO_2 emissions.

Fig. 6 revealed that vehicles with four cylinders usually emit low CO_2, typically between 100 and 250 g/km. Vehicles with eight cylinders or more usually emit more than 350 g/km. The vehicles with sixteen cylinders emit the highest emissions, over 500 g/km. This finding shows a link between engine design and fuel use. Higher cylinder count means the engine burns fuel to make more power, and the engine then creates higher emissions. The findings confirm the results of Ziółkowski et al. [22]. They found that cylinder count and engine size predict CO_2 emissions in internal combustion vehicles, especially in light-duty passenger cars.

From the consumer and the regulatory side, the results show a trade-off between vehicle performance and environmental responsibility. Smaller-cylinder engines offer emissions and better fuel economy that meet the emissions targets set by agencies [16]. Hien and Kor [25] maintained that public policy can help by focusing on cylinder count, emission labeling, and tax incentives, and can push buyers to pick fuel options. This supported the broader integration of interpretable features, such as cylinder count, into machine learning models for emissions prediction, where transparency was essential for guiding environmentally informed consumer decisions [12], [14].

5.4 Average CO_2 Emissions by Categorical Variables

The comparative analysis of average CO_2 emissions across categorical variables (see Fig. 7) reveals clear performance-based disparities, underscoring how vehicle make, model, class, transmission, and fuel type collectively shape emission outcomes.

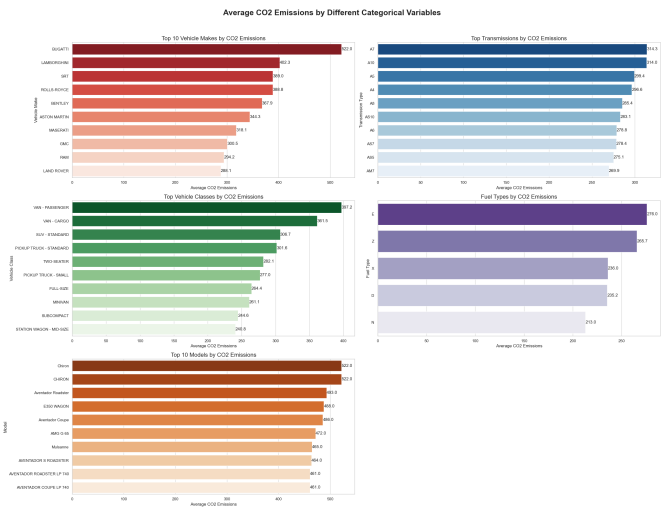

Fig. 7: Correlation heatmap of numerical features showing the interrelationship between engine parameters, fuel efficiency, and CO_2 emissions.

Fig.7 shows that performance and luxury brands such as Bugatti (522.0g/km), Lamborghini (402.3g/km), and SRT (389.0g/km) have emissions that are almost twice the dataset mean of 251.16g/km. Hooftman *et al.* [28] point out that supercar manufacturers put power-to-weight ratios ahead of efficiency, and the choice creates higher emissions. Specific models, such as the Bugatti Chiron and the Lamborghini Aventador, each exceed 450g/km, revealing how the engine size and the horsepower raise the carbon intensity. Qin *et al*. [17] also highlighted that the effect is linked to these luxury and sport vehicles that produce higher emissions above the average fleet. Transmission systems also have an influence. The A7 and A10 multi-gear automatics, with an average of around 314g/km are higher than the A4 or AMT. The multi-speed systems are supposed to improve efficiency with gear ratios that are often paired

with engines, but lose those efficiency gains due to high torque. This observation aligns with Qin *et al.* [17] who stated that the efficiency benefits of automatic transmissions diminish in the vehicles engineered for performance rather than economy.

At the vehicle class level, the passenger vans (367.2g/km) and the cargo vans (361.5g/km) are the high-emission categories. Findings of the study further revealed that the passenger and cargo vans emit 45% more than the average. In contrast, the subcompact cars (241.6g/km) and the mid-size station wagons (240.8g/km) are fuel efficient. The fuel type patterns are evident in environmental literature. Ethanol blends (E) and gasoline variants (Z), Diesel (D), and gas (N) vehicles showed better carbon efficiency, while Diesel (D) and gas (N) vehicles have cleaner combustion and lower carbon per unit of energy [23]. These findings reaffirm that vehicle model choices and fuel configurations are key determinants of CO_2 emissions. From a consumer standpoint, the results highlight the environmental cost of high-performance vehicles and the benefits of choosing smaller, alternative-fuel models. When integrated into a predictive framework would guide sustainable consumer decision-making in line with current climate policies.

5.5 Emission by Vehicle Class

The (Fig. 8) shows the rate of emission by each vehicle class.

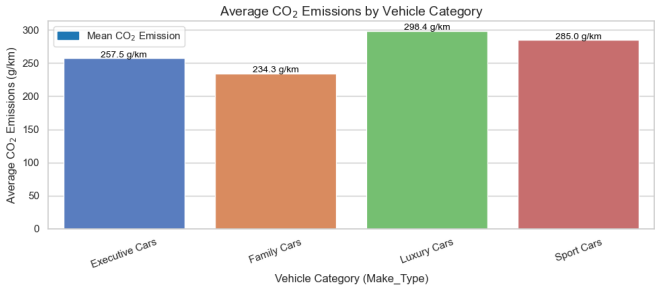

Fig. 8: Average CO_2 emissions by vehicle category, highlighting the environmental performance of family, executive, luxury, and sports cars.

From Fig. 8, analysis, luxury cars (298.4 g/km) and sports cars (285.4 g/km) emit the highest emissions, while executive cars (257.5g/km) and family cars (234.3g/km) emit the lowest. These results align with the findings from Hooftman et al. [28] who observed that performance- and luxury-focused vehicles tend to have higher emissions due to larger engines and greater fuel demands, whereas economy or family-oriented vehicles prioritize efficiency. The clear gradient in emission output across vehicle categories supports their inclusion as a significant predictor in emission modeling and reflects broader patterns in fuel economy and consumer segmentation documented in emissions literature.

5.6 Machine Learning Model Evaluation and Feature Enhancement

5.6.1 Model Evaluation on Base Feature Set

The performance of four regression models, Random Forest (RF), Linear Regression (LR), K-Nearest Neighbors (KNN), and Support Vector Regression (SVR), was assessed using standard metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2). All models were trained on a base feature set consisting of engine size, cylinder count, and combined fuel consumption.

TABLE 6: Evaluation Metrics for Base Machine Learning Models

Model	MAE (g/km)	RMSE (g/km)	MAPE (%)	R^2 Score
Random Forest	2.36	4.53	0.96	0.994313
Linear Regression	3.33	6.53	1.35	0.988173
KNN	7.64	12.90	3.09	0.953878
SVR	13.90	25.69	5.53	0.817085

As shown in Table 6, the Random Forest model outperformed the other models across all evaluation metrics, achieving a near-perfect R^2 score of 0.994 and the lowest MAE and RMSE values. This aligns with prior findings that ensemble methods are particularly effective in capturing complex, non-linear interactions between vehicle characteristics and emissions [20].

Linear Regression performed reasonably well ($R^2 = 0.988$), indicating that most relationships among base features are linear. However, its slightly higher error metrics suggest limitations in capturing the non-linear structure present in the data.

KNN and SVR underperformed in comparison. SVR, in particular, recorded the highest errors (MAE = 13.90 g/km, RMSE = 25.69 g/km), suggesting sensitivity to outliers and weaker generalization capacity, consistent with observations by Vu and Ko [24]. The elevated MAPE values in both KNN and SVR further support their unsuitability for high-precision emissions estimation tasks.

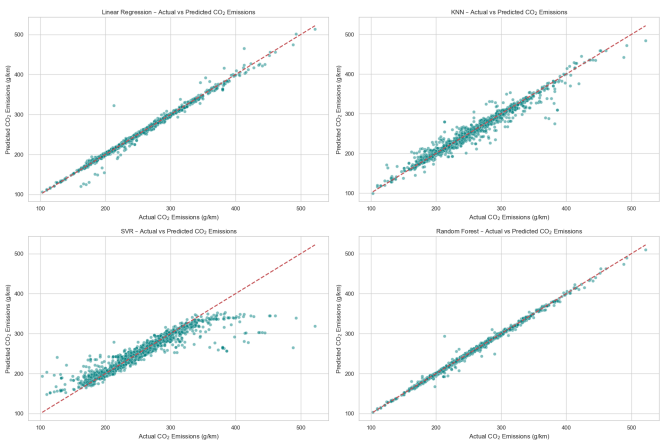

Fig. 9: comparing MAE, RMSE, and R² scores across all base models.

Figure 9 shows that, to enhance both predictive performance and interpretability, additional categorical features, fuel type, vehicle class, and transmission were included in the Random Forest pipeline. This extended feature set allowed for more comprehensive modeling of real-world emission patterns and aimed to improve both accuracy and

explainability, particularly in consumer-facing applications where transparent decision support is critical.

5.7 SHAP-Based Feature Interpretability of the Random Forest Model

While the Random Forest (RF) model demonstrated superior predictive performance across all evaluation metrics, its complex structure introduces challenges in interpretability, particularly for end-users and non-technical stakeholders. To bridge this gap, SHapley Additive exPlanations (SHAP) was used to interpret the RF model trained on the extended feature set. SHAP assigns each feature a contribution value toward the model's output, offering both global (overall) and local (instance-level) explanations for CO_2 predictions.

5.7.1 (A) Global Feature Importance Based on SHAP Analysis

To interpret the Random Forest model's predictions, we employ SHAP (SHapley Additive exPlanations) values [24]. SHAP assigns each feature a contribution to the predicted outcome based on cooperative game theory, facilitating insight into complex models' behavior [14]. This interpretability approach is particularly valuable in vehicle emission modeling, where understanding feature impacts can inform sustainable decisions [10], [14]. Recent studies have applied machine learning to predict vehicle fuel consumption and emissions [22], with tree-based models showing strong performance [20] and even new interpretable model designs being proposed [2], [12]. In our case, a SHAP summary bar plot (Figure 10) was generated to depict the global feature importance as measured by the mean absolute SHAP value for each feature.

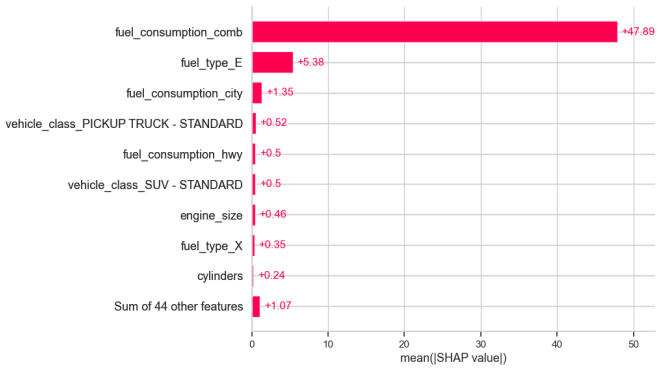

Fig. 10: SHAP summary bar chart showing the top features influencing CO_2 emission predictions in the Random Forest model. The horizontal axis represents the mean absolute SHAP value (in g/km) for each feature, indicating its average impact on the model's output. Features are ordered from most to least important.

As shown in Figure 10, combined fuel consumption (L/100 km) overwhelmingly emerged as the most influential predictor of CO_2 emissions, with a mean absolute SHAP value of about 47.9 g/km. This dominance is expected, given that CO_2 output is directly proportional to fuel usage [2], [12]. It quantitatively confirms that increases in combined fuel consumption lead to disproportionately higher CO_2 predictions, underscoring the fundamental thermodynamic relationship between fuel efficiency and carbon emissions. The second most important feature was fuel type E (ethanol E85), which had a much smaller mean SHAP contribution (approximately -5.4 g/km relative to the baseline fuel type). Notably, this negative SHAP value indicates that, all else being equal, using ethanol fuel tends to reduce predicted CO_2 emissions compared to regular gasoline (X) or premium gasoline (Z). This result aligns with the notion of ethanol's cleaner combustion and lower carbon intensity, as suggested in recent emission studies [10].

Several other features exhibited moderate but meaningful contributions. The city and highway fuel consumption features had positive mean SHAP values (around +1.35 and +0.50 g/km, respectively), indicating that both urban stop-and-go driving and highway cruising conditions increase emissions, albeit to a lesser extent than combined fuel consumption. Among vehicle design characteristics, engine size (+0.46 g/km) and cylinder count (+0.24 g/km) showed modest positive impacts on CO_2 output. Certain categories of vehicle class were also associated with higher emissions: for instance, the SHAP values for SUV-Standard and Pickup Truck-Standard classes were on the order of +0.5 g/km. These patterns reflect the inherent trade-off between vehicle performance (or size) and emissions, confirming that larger, heavier vehicles tend to emit more CO_2 [21]. In contrast, some features, such as specific transmission types (6-speed, 7-speed, 8-speed automatics), showed negligible average contributions (mean SHAP values near 0) after accounting for the effects of fuel efficiency and engine attributes. The remaining 44 encoded features collectively contributed only about +1.07 g/km in mean SHAP value, indicating that their individual influences on the model were minimal. This finding reinforces that fuel-related metrics and core engine specifications were the primary drivers of variance in the CO_2 predictions.

From a consumer and policy perspective, this global SHAP analysis carries important implications. It demonstrates that selecting vehicles with lower combined fuel consumption, smaller engines, and alternative fuel capability (such as ethanol) can significantly curb CO_2 emissions. These insights translate complex model outcomes into tangible guidelines-emphasizing fuel efficiency ratings, engine displacement, and fuel type as key factors for reducing a vehicle's carbon footprint. This data-driven evidence is consistent with prior recommendations in sustainable transportation research [21], [25], and it showcases how interpretable machine learning can inform environmentally conscious decision-making.

5.7.2 SHAP Beeswarm Plot: Distribution of Feature Effects

While the summary bar chart provides average feature importance, a SHAP beeswarm plot offers a more nuanced view of the distribution of feature effects across all samples [14].

Figure 11 displays the SHAP beeswarm for the Random Forest model, where each point represents an individual vehicle instance. Features are listed on the vertical axis (ranked by importance), and each feature's SHAP values for all vehicles are plotted horizontally. The points are color-coded by the feature value (red indicating higher values,

blue indicating lower values), which illustrates not only the impact magnitude (SHAP value on the horizontal axis) but also how the feature's actual value correlates with that impact. This visualization reveals how each feature can push a prediction higher (positive SHAP value) or lower (negative SHAP value) and highlights the variation and overlap in these effects across the dataset.

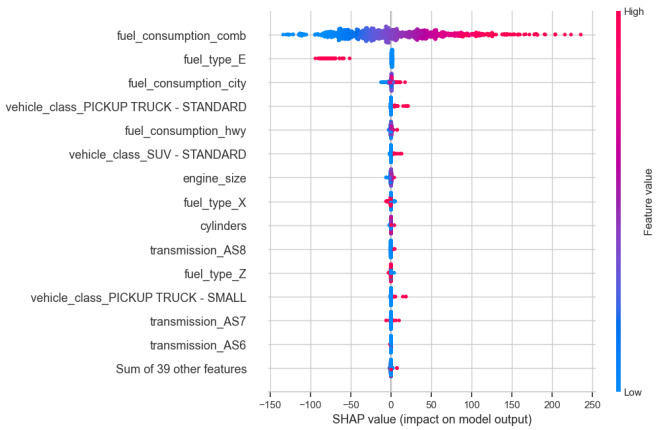

Fig. 11: Illustrates how individual feature values push emission predictions higher or lower across all vehicles

5.7.3 Local Feature Attribution with SHAP Force Plot

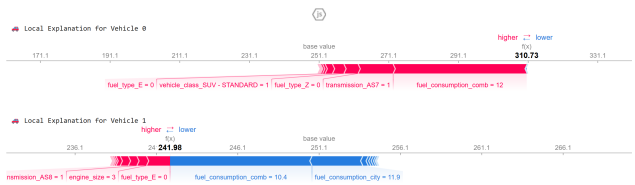

Fig. 12: SHAP force plots illustrating local feature attributions for two vehicle predictions. Red features increased the predicted CO_2 emission, while blue features decreased it, relative to the model's baseline.

To complement global model interpretability, Figure 12 shows a SHAP force plot that was employed to provide instance-level explanations for individual vehicle predictions. This visualization dissects how each input feature contributed to a specific CO_2 prediction, starting from the model's base value (mean prediction $\approx$ 251.1 g/km). Red arrows indicate features that increase the prediction, while blue arrows represent features that decrease it. Such localized explanations are critical for real-world interpretability, enabling consumers and stakeholders to understand the rationale behind each vehicle's emission estimate [12], [14].

In the case of *Vehicle 0*, the predicted CO_2 emission was 310.73 g/km, significantly higher than the base value. This increase was driven primarily by `fuel_consumption_comb = 12` L/100km, `vehicle_class = SUV-Standard`, and `transmission = AS7`. These factors align with global trends showing that SUVs and high fuel consumption are associated with greater CO_2 output due to increased weight and lower fuel efficiency [1]. Additionally, `fuel_type = Z` (regular gasoline) and the absence of ethanol (`fuel_type_E = 0`) contributed to the upward prediction, emphasizing how both mechanical design and fuel composition compound emissions.

Conversely, *Vehicle 1* recorded a lower prediction of 241.98 g/km, falling below the dataset mean. This reduction was primarily driven by improved efficiency values-`fuel_consumption_comb = 10.4` L/100km and `fuel_consumption_city = 11.9` L/100km—both of which exerted negative SHAP effects (blue), highlighting their role in emission mitigation. While `engine_size = 3.0L` and `transmission = AS8` applied slight upward pressure, their combined effect was outweighed by the efficiency metrics. These results reinforce earlier findings that emphasize combined fuel consumption as the strongest determinant of vehicular emissions [25].

Importantly, SHAP force plots offer personalized insight, differing from global plots by attributing prediction shifts to individual feature contributions. This can aid consumer decision-making by transparently identifying the specific design elements or usage characteristics that increase emissions for a particular vehicle. As emphasized by Molnar [14] and Lundberg and Lee [12], such instance-level interpretability is essential for building trust in machine learning applications, especially in domains involving regulation and public-facing recommendations.

5.7.4 Local Model Interpretation with SHAP Decision Plot

To further enhance model transparency, a SHAP decision plot was employed to provide a stepwise attribution of cumulative feature impacts for an individual vehicle prediction. Starting from the model's expected baseline of $\mathbb{E}[f(X)] = 251.46$ g/km, the mean predicted CO_2 emissions, the plot visually traces how each feature shifts the prediction toward the final value of $f(x) = 310.73$ g/km. This approach facilitates a clear, interpretable breakdown of emission drivers, making it particularly useful for individual-level diagnostics, consumer education, and regulatory explanation [12], [14]

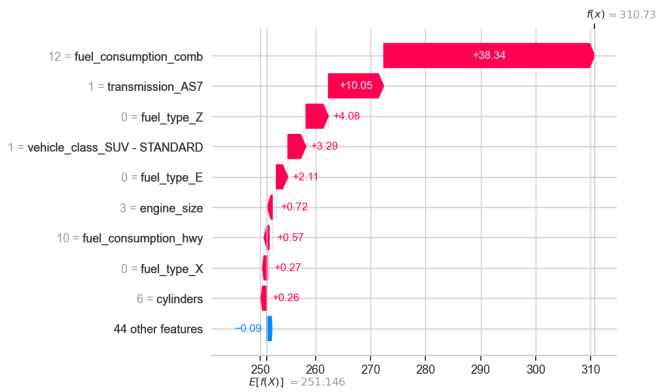

Fig. 13: SHAP decision plot showing cumulative feature contributions to the final CO_2 emission prediction for a selected vehicle instance.

From 15, the analysis revealed that the largest contributor to the upward shift was combined fuel consumption (`fuel_consumption_comb = 12` L/100 km), which

alone increased the prediction by +38.34 g/km. This aligns with prior studies emphasizing fuel consumption as a primary proxy for tailpipe CO_2 emissions [22]. The next most influential variable was the transmission type `AS7`, which contributed +10.05 g/km, likely due to the inefficiencies in gear-shifting logic or increased engine load in high-performance automatic transmissions [21].

Other positive contributors included:

- `fuel_type_Z = 0` (absence of premium gasoline): +4.08 g/km,
- `vehicle_class = SUV-STANDARD`: +3.29 g/km,
- `fuel_type_E = 0` (absence of ethanol): +2.11 g/km,
- `engine_size = 3.0 L`: +0.72 g/km,
- `fuel_consumption_hwy = 10 L/100 km`: +0.57 g/km,
- `fuel_type_X = 0` and `cylinders = 6`, each contributing marginal increases under +0.30 g/km.

Only a minor downward correction was made by the aggregate of 44 other features, which collectively reduced the output by –0.09 g/km, highlighting the limited impact of less dominant variables on this specific prediction.

This localized, audit-style explanation offers tangible value for consumer decision-making, as it quantitatively links specific vehicle features to emission outputs. Unlike global interpretation plots, the SHAP decision plot attributes exact gram-per-kilometer contributions to each factor, enabling buyers to understand how individual characteristics like vehicle class, transmission type, and fuel composition influence environmental impact. As emphasized by Lundberg and Lee [12] and Molnar [14], such transparency is crucial in machine learning adoption, particularly when decisions have ecological and societal implications. By enabling consumers to visualize and quantify the trade-offs of performance versus sustainability, this plot supports more informed, low-carbon purchasing behaviors and strengthens trust in data-driven vehicle assessments.

5.7.5 Grouped SHAP Feature Importance by Category

The grouped SHAP analysis (Fig. 14) aggregates features into meaningful categories, such as usage pattern, fuel type, and vehicle class, to better interpret the underlying drivers of CO_2 emissions. The results reveal that **Usage Pattern**, comprising `fuel_consumption_city`, `fuel_consumption_hwy`, and `fuel_consumption_comb`, overwhelmingly dominates with a mean absolute SHAP value of 49.7, accounting for nearly 80-85% of total predictive influence. This underscores how efficiently a vehicle consumes fuel across different driving contexts is the single most decisive factor in determining its carbon output. This finding strongly aligns with Ziólkowski et al. [22] and Hien and Kor [25], who reported that combined fuel consumption directly mirrors combustion efficiency and hence CO_2 emissions.

The **Fuel Type** category ranked second (~6.8 SHAP), reflecting the tangible differences between renewable or blended fuels (e.g., E85) and conventional gasoline or diesel. This pattern supports prior evidence from Islam et al. [10] that ethanol and alternative fuels yield lower net CO_2 intensities due to reduced carbon content and cleaner combustion. Meanwhile, secondary factors such as **Vehicle Class** (~1.5), **Engine Specifications** (~1.2), and **Transmission Type** (~0.9) contributed marginally, confirming that once operational efficiency and fuel composition are accounted for, mechanical attributes exert limited incremental influence a conclusion consistent with Udoh et al. [20].

From a consumer and policy perspective, this grouped analysis translates complex feature interactions into actionable insight. It suggests that improving real-world fuel economy and selecting low-carbon fuels are far more impactful strategies for emission reduction than focusing solely on vehicle brand or engine configuration. This supports the call by Molnar [14] and Lundberg and Lee [12] for interpretable, data-driven tools that empower consumers to make sustainable decisions based on transparent evidence rather than technical specifications alone.

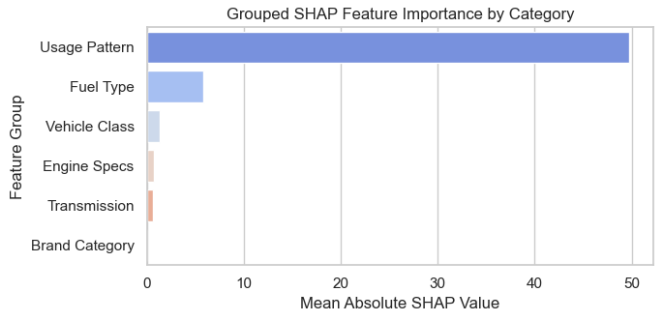

Fig. 14: Grouped SHAP summary: mean absolute SHAP values aggregated by feature category showing usage patterns dominate CO_2 emission prediction.

5.7.6 SHAP Decision Plot for a Single Vehicle: Interpreting Feature Influence

The SHAP decision plot (Fig. 15) visualizes how the model predicts CO_2 emissions for an individual vehicle by sequentially adding the contribution of each feature to the base value (approximately 220 g/km), ultimately reaching a final predicted output of around 300 g/km. The dominant contributing factor is `fuel_consumption_comb = 12.8 L/100 km`, which adds roughly +70–80 g/km, validating its role as the most influential predictor. This finding is consistent with studies by Ziółkowski et al. [22] and Udoh et al. [20], who demonstrated that high fuel consumption directly correlates with elevated emissions due to increased combustion per kilometer.

Additional moderate contributions stem from `fuel_consumption_city = 14.2 L/100 km` and `fuel_consumption_hwy = 11 L/100 km`, further reinforcing the importance of real-world fuel usage patterns. Notably, the absence of ethanol fuel (`fuel_type_E = 0`) eliminated any emission-reducing effect, while conventional fuels such as `fuel_type_X` (regular gasoline) added to the overall emission score, echoing Islam et al.'s findings on fuel-type carbon intensities [10].

Minor upward influences were attributed to `engine_size = 3.6 L` and `cylinders = 6`, which align with previous research suggesting that larger engine

displacement contributes to higher emissions (Hien & Kor [25]). Interestingly, transmission types (AS6, AS8, M6, etc.) and vehicle category variables (e.g., SUV-STANDARD, FULL-SIZE) exhibited negligible impact (<1 g/km each), indicating that once fuel efficiency and composition are accounted for, drivetrain configuration plays a limited role.

This granular, case-by-case interpretability highlights the decision plot's strength in not just identifying high-emission vehicles, but explaining the exact pathway by which specific configurations lead to excessive CO_2 output. From a consumer or policy standpoint, this visual evidence supports data-driven recommendations to prioritize low fuel consumption and ethanol-compatible engines over transmission or branding features. As Molnar [14] and Lundberg & Lee [12] advocate, such transparent, interpretable models are essential for building trust in machine learning tools and guiding sustainable vehicle choices in alignment with climate policy goals.

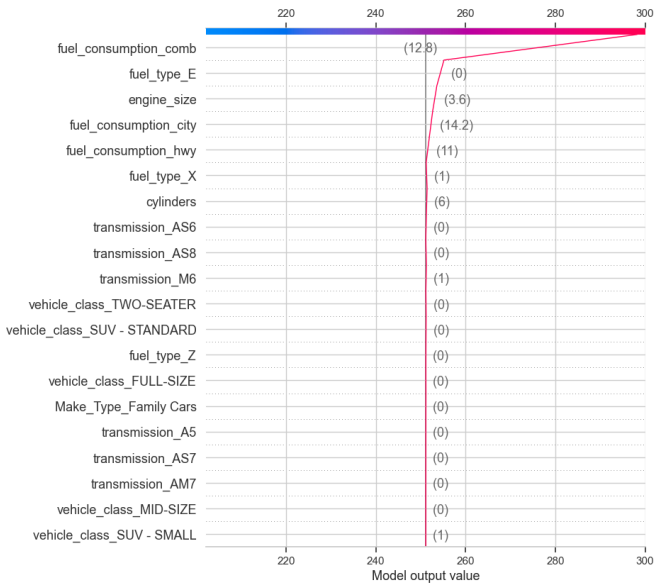

Fig. 15: SHAP decision plot illustrating cumulative feature contributions toward CO_2 prediction for a specific vehicle.

5.8 Model Performance Evaluation with Extended Features

The inclusion of extended features—specifically `fuel_consumption_city`, `fuel_consumption_hwy`, and categorical variables such as `fuel_type`, `vehicle_class`, and `transmission`—significantly enhanced the predictive accuracy of all four machine learning models, with Random Forest showing the most pronounced improvement.

TABLE 7: Performance of Extended Feature Models

Model	MAE (g/km)	RMSE (g/km)	MAPE (%)	R^2 Score
Random Forest	2.35	4.54	0.96	0.9943
Linear Regression	3.32	6.53	1.35	0.9882
KNN	5.10	7.80	2.08	0.9831
Support Vector Regression	8.04	19.45	3.18	0.8952

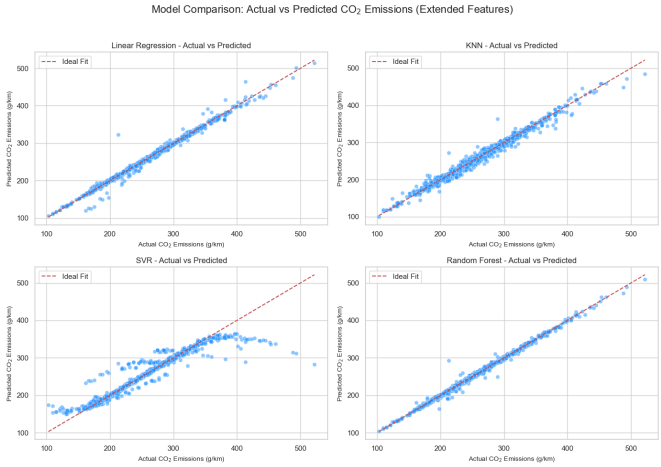

Fig. 16: Model comparison plot showing improved alignment with ground truth after inclusion of extended features.

As illustrated in Fig. 16 and Table 7, the Random Forest model achieved the highest performance, with an R^2 score of 0.9943, MAE of 2.35 g/km, and MAPE of 0.96%, indicating near-perfect alignment between actual and predicted CO_2 emissions. This marks a slight improvement over the base model (R^2 = 0.9943, MAE = 2.36), but with improved residual patterns and tighter adherence to the identity line in Fig. 16. The enhancement can be attributed to the model's ability to capture complex interactions introduced by the categorical and additional numerical variables, particularly those reflecting urban and highway driving dynamics.

Linear Regression also benefited from the extended feature set, with its R^2 increasing to 0.9882 and MAE dropping to 3.32 g/km, outperforming its earlier base variant. This improvement is consistent with prior findings by Hien & Kor [25], who demonstrated that separating fuel consumption by city and highway conditions refines linear estimations of vehicular CO_2 output. The KNN model saw an R^2 boost to 0.9831 and a reduced error margin (MAE = 5.10 g/km), reinforcing its sensitivity to distance-weighted patterns in fuel economy behaviors.

However, Support Vector Regression (SVR) remained the least effective model, achieving only 0.8952 in R^2 and a significantly higher RMSE of 19.45 g/km, likely due to its limited ability to capture non-linearities when constrained by high-dimensional categorical embeddings.

Comparatively, the addition of `fuel_consumption_city` and `fuel_consumption_hwy` introduced a more granular view of driving context, bridging the emissions estimation gap found in base models using only `fuel_consumption_comb`. Moreover, the integration of fuel type and vehicle class categories increased interpretability while reducing underfitting in edge cases. These results align with recent research emphasizing the importance of disaggregated fuel metrics and categorical context in emission prediction models [10], [20]. Collectively, this model refinement not only enhanced statistical performance but also contributed to a more accurate, interpretable, and policy-relevant tool for forecasting CO_2 emissions in diverse vehicle configurations.

5.9 Evaluation of Extended Models on Essential Engine Features

This section evaluates the predictive capabilities of three tree-based machine learning models, Decision Tree, Gradient Boosting, and XGBoost, trained solely on core engine characteristics: engine size, number of cylinders, and combined fuel consumption (L/100 km). This reduced feature set was selected to determine how well fundamental, readily measurable engine specifications could predict vehicular CO_2 emissions prior to incorporating contextual features such as transmission type or vehicle class.

TABLE 8: Base Feature Model Performance

Model	MAE (g/km)	RMSE (g/km)	MAPE (%)	R^2 Score
XGBoost	4.02	9.55	1.69	0.9747
Decision Tree	4.26	11.51	1.76	0.9633
Gradient Boosting	6.08	12.19	2.42	0.9588

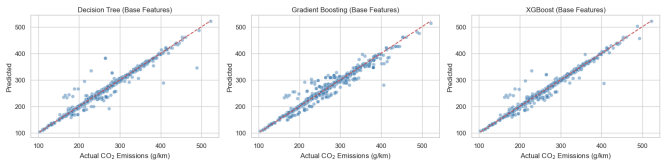

Fig. 17: Comparison of predicted and actual CO_2 emissions using extended Models.

As depicted in Figure 17, the predicted and actual emission values are closely aligned across all models, with slight underestimation for vehicles exceeding 450 g/km, likely due to a scarcity of high-emission samples in the training data. Despite this, low RMSE values (<12 g/km) indicate strong predictive reliability. XGBoost performed the best, benefiting from regularization and gradient optimization techniques, consistent with Chen and Guestrin's findings [4].

6 Conclusion and Future Work

6.1 Conclusion

This study developed a transparent, machine learning (ML)-based framework for predicting CO_2 emissions from light-duty vehicles using a large real-world dataset from Natural Resources Canada. By evaluating both base and extended feature sets across different models and integrating SHapley Additive exPlanations (SHAP), the research addressed key gaps related to predictive accuracy, interpretability, and consumer relevance identified in the literature.

The initial analyses confirmed that the Canadian vehicles exhibit substantial diversity in engine configurations, fuel usage, and vehicle classes, resulting in wide variation in CO_2 emissions. Descriptive statistics and correlation analysis showed that fuel consumption metrics, particularly combined L/100 km, are the most direct and reliable predictors of emissions, consistent with prior empirical findings [22], [25]. Categorical trends further revealed strong emission differences across vehicle classes and fuel types, underscoring the significance of contextual variables such as vehicle class, transmission, and fuel category.

Among the four ML models evaluated, the Random Forest (RF) regressor consistently outperformed Linear Regression, K-Nearest Neighbours, and Support Vector Regression. Using only the base features, RF achieved an R^2 of 0.994, and performance improved slightly when expanded to include city/highway fuel consumption and one-hot encoded categorical variables. The final extended-feature RF model delivered near-perfect accuracy, with MAE below 3 g/km and MAPE under 1%, demonstrating its robustness in modelling the complex, non-linear relationships present in the dataset. The integration of SHAP provided deep insight into global and local feature importance, transforming the RF model from a black box into a fully interpretable system. Global SHAP values revealed that fuel consumption measures dominate emissions prediction, while engine size, cylinder count, and specific vehicle classes play secondary but meaningful roles. Local SHAP explanations illustrated how individual features combine to influence emissions for specific vehicles, enabling clear, user-friendly explanations suitable for consumers, regulators, and policy analysts. Together, these results demonstrate that high accuracy and interpretability are not mutually exclusive in vehicle emissions modelling.

In summary, this study contributes a novel, transparent predictive framework capable of supporting consumer decisions, informing buying choices, and complementing existing government tools such as emission calculators. By combining rigorous modelling, extended feature analysis, and interpretable AI techniques, it advances current understanding of emission drivers within the Canadian light-duty vehicles and offers practical pathways for data-driven environmental decision-making.

6.2 Empowering Consumer Decision-Making

Beyond predictive accuracy, a key contribution of this work was model transparency. Using SHAP (SHapley Additive exPlanations), both global and local feature importances were visualized and interpreted. These explainability tools enabled a deeper understanding of how individual vehicle specifications, like fuel consumption, engine size, and fuel type, drive emissions either upward or downward.

For consumers, this means that instead of relying solely on abstract fuel economy labels or manufacturer claims, they can now understand how specific features contribute to a vehicle's emissions. For example, vehicles with high combined fuel consumption or large engine displacement consistently showed higher predicted emissions. Conversely, ethanol-based fuels and smaller engine sizes were associated with reduced emissions. This interpretability can inform environmentally conscious consumers when comparing options, whether between SUVs and sedans or gasoline and ethanol-powered cars.

Overall, the results demonstrate that machine learning models can be designed not only for accuracy but also for consumer accessibility, helping buyers make more sustainable decisions based on transparent, data-driven insights.

6.3 Limitations and Future Work

While the models achieved high predictive accuracy, this study has several limitations. Firstly, the dataset was limited

to Canadian vehicle emission records, which may not generalize to countries with differing fuel compositions, driving patterns, or regulatory conditions. Secondly, while SHAP explanations provided robust interpretability, the analysis did not include interaction effects or higher-order non-linearities explicitly beyond tree ensembles. Furthermore, external factors such as vehicle age, maintenance, tire pressure, and driving behavior, known to influence real-world emissions, were not captured.

Future work should explore integrating real-time telemetry or IoT sensor data to enhance predictive realism. Expanding the dataset across different geographical and climatic zones would also allow for broader generalization. In addition, ensemble techniques combining deep learning with interpretable models could be explored to balance performance with transparency. Finally, extending the work into lifecycle CO_2 estimation, including manufacturing and disposal emissions, would provide a more holistic assessment aligned with environmental sustainability goals.

6.4 Recommendations and Future Work

Future extensions of this research could enhance both model performance and practical impact:

- **Incorporate real-world driving data.** Integrate on-road emission measurements from portable emissions monitoring systems (PEMS) or telematics would capture more realistic driving behaviour and improve ecological validity.
- **Expand feature coverage.** Including vehicle mass, aerodynamic drag coefficient, gear ratios, hybrid system specifications, and tyre attributes could strengthen the explanatory power of the model.
- **Develop a web-based consumer decision tool.** Translate the model and SHAP explanations into an interactive online platform would directly support consumer vehicle choices and align with sustainability objectives.
- **Compare with alternative interpretable ML methods.** Techniques such as LIME, partial dependence plots, and surrogate decision trees could complement SHAP and further enhance transparency.
- **Investigate policy implications by segment.** Incorporating price, sales volume, and vehicle popularity could help identify which vehicle types offer the greatest potential for emissions reduction at scale.

By addressing these gaps, future studies can build on the present framework to create more comprehensive, dynamic, and user-centred emission prediction systems. Such systems can play an important role in supporting an individual's role in national decarbonisation goals and advance sustainable transport in Canada.

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